Credit Card Fraud Report

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CSCE 581

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For my project I chose to create a predictive model for financial fraud. This is something customers, developers, law enforcement, and banking institutions all care about. First I plotted out the dataset I used (https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud) and plotted out the data:

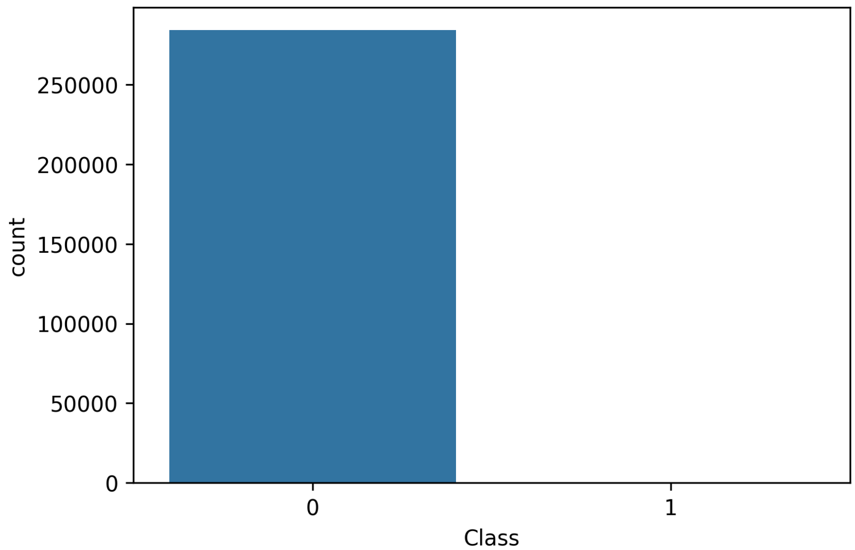
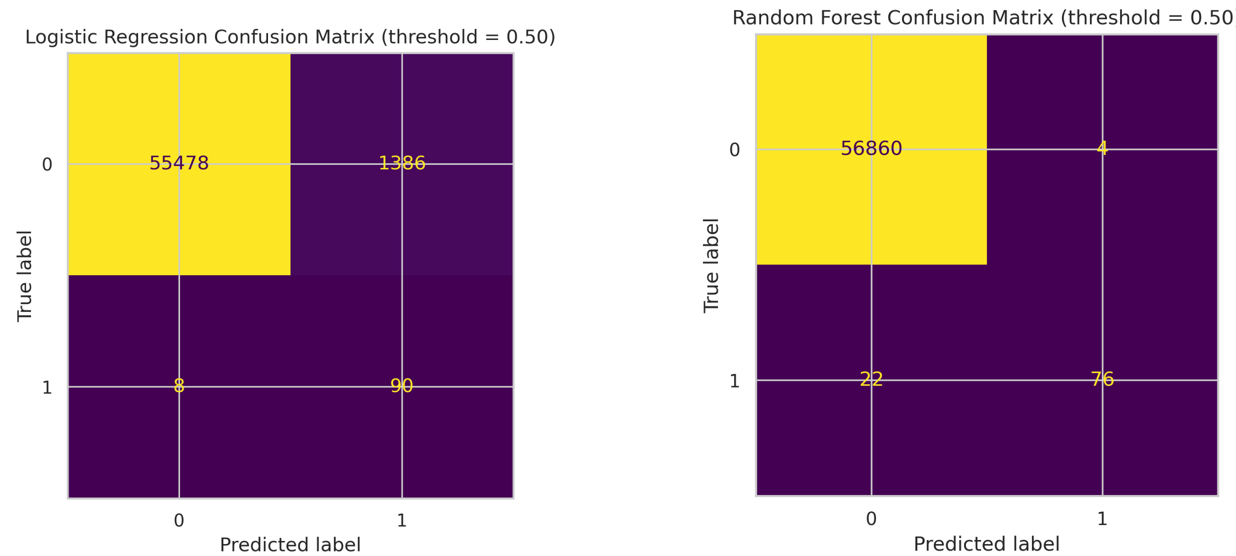


Figure 1: Count signifies the number of parameters. A class of 0 signifies a legitimate transaction, and 1 signifies fraud.

Here we have a problem. When I first trained my model, it was much more experienced at identifying legitimate transactions than fraudulent ones because of its heavily imbalanced dataset. As a result, my first models using logistic regression and random forest performed badly at detecting fraud:



(a) Logistic Regression (b) Random Forest

Figure 2: Confusion matrices on default settings

Here logistic regression is flagging too many transactions as fraudulent. When it guesses a transaction is legitimate, it's right 99% of the time but is only correct 11% of the time when it guesses fraudulent. Random forest performs much better, probably due to its ability to capture nonlinear relationships in the data. We can remedy this issue with logistic regression by adjusting the threshold it uses to make a decision; the default threshold is 0.5, which is much too small for such an unbalanced dataset. We want the model to be much more sure than half-sure that a transaction is fraudulent, so I began adjusting the threshold and recording results:

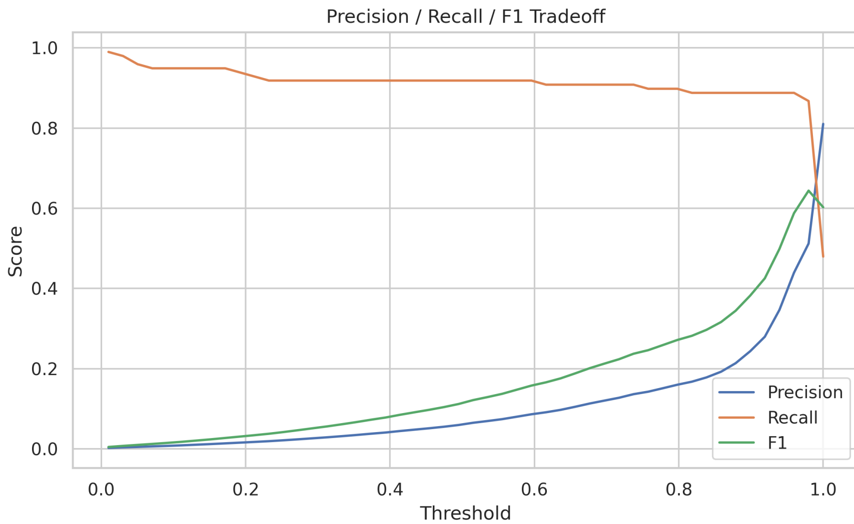


Figure 3: Scores when adjusting threshold

The threshold with the best tradeoff between precision and recall is about 0.98, as indicated by its F1 score. We can achieve much better performance with this adjustment:

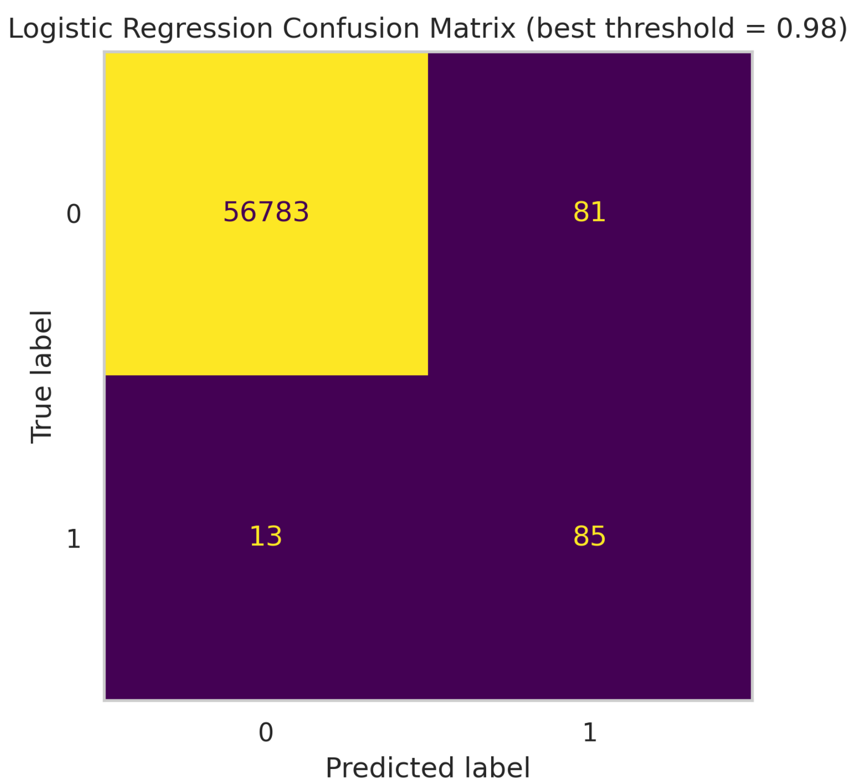
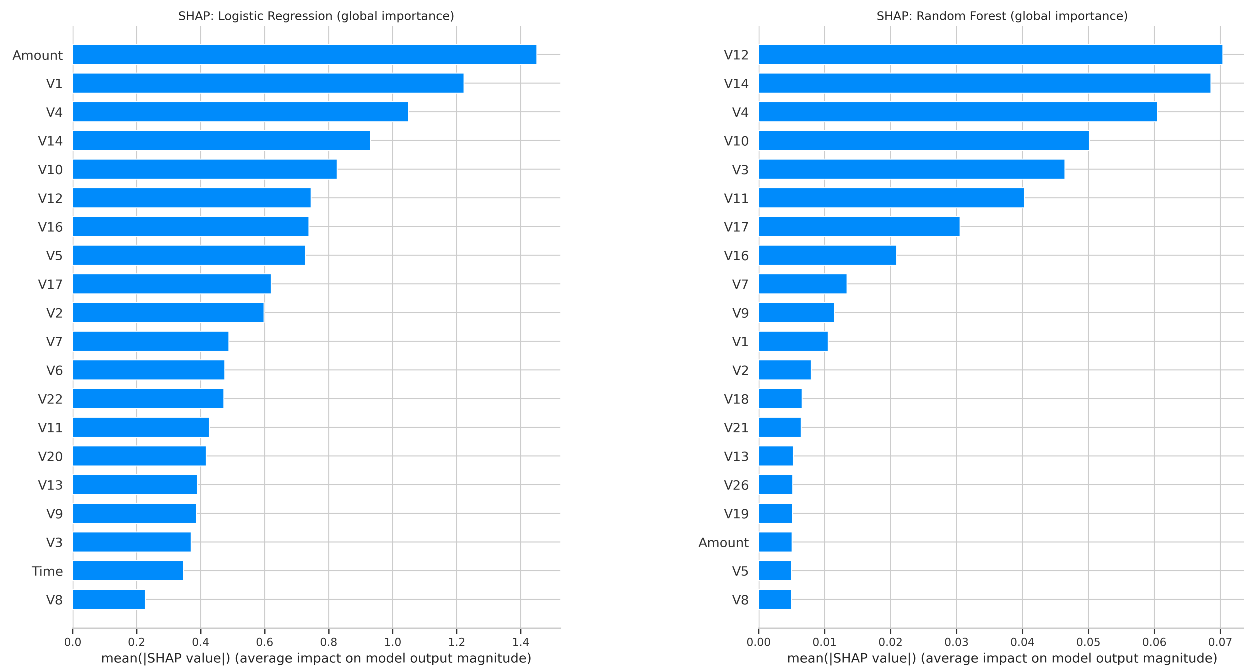


Figure 4: Confusion matrix after setting a threshold of 0.98

After refining my models I investigated what features they were using with SHAP. The features in my dataset were obfuscated to protect customer privacy.



(a) Logistic Regression (b) Random Forest

Figure 5: Most significant features

Interestingly, logistic regression and random forest seem to use somewhat different features to decide. Transaction amount and feature V1 are the two most prominent for logistic regression but are neglected by random forest. I assume this is also partially responsible for random forest's better performance at the default threshold. However, after adjusting the threshold, the logistic regression model's performance is competetive with that of the random forest model.

Doing this project I learned that caution is needed when creating credit fraud models. A model that is too permissive can let financial fraud slip by, while an overly aggressive model can flag legitimate transactions and make banking more difficult. Increasing the decision threshold will increase the amount of false negatives and increasing it will decrease the amount of false positives. This is a fundamental tradeoff in designing decision-making systems. As for the features, transaction amount was the most important feature for logistic regression, which suggests a strong correlation. When user privacy is a concern in datasets, features can be abstracted while still providing the means to create a competent predictive model.